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Introduction to spreadsheets

I. Options:
    A. Microsoft excel
        1. Cut down version of microsoft online
    B. Google sheets
        1. Web based application from google
    C. Libre Office calc
        1. Free and open source desktop spreadsheet application
    D. Zoho sheet, open office calc, quip, smartsheet (project management), apple numbers
II. Advantages
    A. Accurate calculations
    B. aAutomatic calculations
    C. Organize and access data
    D. Format, filter and sort data
    E. Edit, undo, and error-check
    F. Analyze data
    G. Create graphs and other analytics
III. The most common business uses for spreadsheet applications:
    A. Data and entry storage
    B. Comparing large data sets
    C. Modeling and planning
    D. Charting
    E. Identifying trends
    F. Flowcharts
    G. Tracking business sales
    H. Financial forecasting
    I. Stat analysis
    J. Profit and loss accounting
    K. Budgeting
    L. Forensic auditing
    M. Payroll
    N. Personal expenses
    O. Household budgeting
    P. Recipe library
    Q. Fitness tracking
    R. Calorie counting & weigh monitoring
    S. Lists
IV. How a data analyst uses spreadsheets
    A. Collecting data
    B. Cleaning data

C. Analyzing data
D. Visualizing data

SPreadsheets basics part I & 2

I. Workbooks represent an .xlsx file
    A. Contain data, calculations, and analysis
    B. 3D reference is when you refer to cells in another excel workbook
II. Hide
    A. Ctrl+F1
    B. Can add quick shortcuts to quick ribbon including sort ascending and descending
    C. Ctrl+home takes back to cell A1
    D. CTRl+end takes you to end
    E. CTRL+ bottom arrow takes to bottom of sheet and CTRL+up arrow takes you to top
    F. Hold shift+ arrow keys to select multiple data sets
    G. Ctrl+A will select all data and not the cells without data
    H. Select, move, and fill/copy symbol

Reading: Excel Keyboard Shortcuts

Estimated time needed: 30 minutes

The table below lists keyboard shortcuts for some of the most common Excel tasks.

Task	Shortcut
Close a workbook	Ctrl+W

Open a workbook	Ctrl+O
Save a workbook	Ctrl+S
Copy	Ctrl+C
Cut	Ctrl+X
Paste	Ctrl+V
Undo	Ctrl+Z
Remove cell contents	Delete
Bold	Ctrl+B
Open context menu	Shift+F10
Expand or collapse the ribbon	Ctrl+F1
Move up one cell in the worksheet	Up arrow key
Move down one cell in the worksheet	Down arrow key
Move one cell left in the worksheet	Left arrow key
Move one cell right in the worksheet	Right arrow key
Move to the edge of the current data region in the worksheet (e.g. end of column)	Ctrl+Arrow key (e.g. Ctrl+Down arrow)
Move to the last cell on a worksheet	Ctrl+End
Move to the beginning of a worksheet	Ctrl+Home

Extend the selection of cells to the last used cell on a worksheet (lower right corner)	Ctrl+Shift+End
Move to the cell in the upper-left corner of the window (when Scroll Lock is On)	Home+Scroll Lock
Move one screen down in a worksheet	Page Down
Move one screen up in a worksheet	Page Up
Move one screen to the right in a worksheet	Alt+Page Down
Move one screen to the left in a worksheet	Alt+Page Up
Move to the next sheet in a workbook	Ctrl+Page Down
Move to the previous sheet in a workbook	Ctrl+Page Up
Edit the active cell and put the cursor at the end of the cell's contents	F2
Enter the current time	Ctrl+Shift+colon (:)
Enter the current date	Ctrl+semi-colon (;)

Using spreadsheets as a data analysis tool

Summary and Highlights

In this lesson, you have learned:

- There are several spreadsheet applications available in the marketplace; the most commonly used and fully-featured spreadsheet application is Microsoft Excel.
- Spreadsheets provide several advantages over manual calculation methods and they help you keep data organized and easily accessible.
- As a Data Analyst, you can use spreadsheets as a tool for your data analysis tasks.
- There are several elements that make up a workbook in a spreadsheet application.
- The ribbon provides access to all the features and tools required to view, enter, edit, manipulate, clean, and analyze data in Excel.
- There are several ways to navigate around a worksheet and workbook in Excel.

Shortcut:

1. CTRL+F6 to switch between workbooks
2. Ctrl+drag will copy-paste
3. ALT + ??? for sums?

Absolute vs relative:
Relative is referencing the position of the cell, absolute maintains the same position of the cell.

Error codes:

1. #Name= contains unrecognized text. Can get help on that error with info and can show formula. Can choose ignore error and can edit error in formula bar

Summary and Highlights

In this lesson, you have learned:

- There are several features to modify views in Excel, and it is very straightforward to enter and edit data in a spreadsheet.

- You can move or copy data within a worksheet or between worksheets, and you can use AutoFill to automatically enter data that is in a series or that fits a pattern.
- You can format both cells and data in Excel.
- A formula is made up of several component parts, and formulas can perform calculations using numbers directly or by using references to data in the worksheet.
- You can use the Fill Handle in Excel to quickly copy formulas to other cells.
- There are several different categories of function you can use for different purposes, and you can search for a function by name, or by category.
- You can reference cells in the worksheet in your formulas by using relative, absolute, or mixed references.
- You can make a formula absolute by adding a dollar symbol ($) to a cell reference.
- If you get errors in your formulas, you can use the error-checking capabilities of Excel to resolve them.

5 traits of good data:
Accuracy, completeness, timeliness, reliability, and relevance

I. Accuracy is first and most significant aspect to data quality. Must remove blanks
II. See if dataset is complete and readily available
III. Reliability - maybe data is unreliable
IV. Relevance.
    A. Is data really necessary for the project?
V. Timeliness
    A. Availability and accessibility of data

Importing file data

I. Plain test, comma or tab separated files can also be used and imported into excel.
II. Text import wizard.
    A. FIle>open>select file to import. Will have text import wizard that will try to figure out what the file is
    B. Select my data has headers
    C. Select delimiter (comma for csv, etc)
    D. Set data format if you feel like it

Fundamentals of Data privacy

I. Confidentiality, collection and use, compliance:
    A. Confidentiality is an acknowledges that customer's personal information belongs to them
    B. Personal information (PI) is any information that can be traced to to that specific individual (emails to images)
    C. Personally identifiable information or PII is specific information that could be used to identify an individual (ss number, DL number)
    D. Sensitive personal information may not identify an individual but is private info that could be used to harm the individual (race, sexual orientation, biometric info, etc)
    E. Can remove unnecessary information
        1. General data protection regulatorion (GDPR) is regulation specific to Europe
        2. LGPD will take effect in 2020 in Brazil and ignores location data process
        3. US doesn't have one law, but california has consumer privacy act to better protect data
        4. Also industry specific regulations such as HIPAA that govern collection and disclosure of protected health information,
        5. PCI standards govern credit card
        6. Understanding data privacy regulations of EU, US, and other countries is key to keeping data safe.
    F. Data privacy applies to most data that is collected, there are some instances where they don't apply, particular collection of data must be completely anonymous

Summary and Highlights

In this lesson, you have learned the following information:

The Five Traits of Data Quality:

- Accuracy
- Completeness
- Reliability
- Relevance
- Timeliness

Importing Text:

- You can use the ‘Text Import Wizard’ to import data from other formats, such as plain text, or comma-separated value files.

The Three Fundamentals of Data Privacy:

- Confidentiality
- Collection and Use
- Compliance

Removing duplicated or inaccurate data and empty rows

I. Check for spelling
    A. Select data> spelling (review tab in ribbon)
II. Empty rows
    A. Select whole data set: Ctrl+Shift+end
    B. Select filter icon and uncheck select all, check blanks
    C. Select these rows and delete the rows. Remove filter and re-view
III. Duplicated rows of data
    A. Select column of data you wouldn’t expect to have duplicate data like sales b/c far less likely these numbers will be duplicated
        1. Conditional format>highlight duplicate values
        2. Delete any info that is dupliacated
    B. Simpler but less secure method
        1. Select whole data sheet, on data tab use remove duplicates button, select only sales column and it will show you the number of deletions
IV. Find and replace
    A. Under home tab
    B. Can find and replace all instances if you want

Dealing with inconsistencies in Data

I. Functions:
    A. Upper, lower, and proper functions to change case in data
        1. Insert a “helper” row, then =proper(A1) and enter
        2. Shift+ right arrow, then ctrl+enter to past the functions
            a) Then need to paste values
    B. Upper to change all to upper case
    C. Lower to change all to lower case
    D. Date format
        1. Change local to english/US from format cells ribbon
        2. Can also do custom format
    E. White space
        1. Use find and replace

a) Select all data
b) Home tab find and select
c) Enter double space and replace with single space
(1) Could do replace all but don’t do that, check it
2. Use trim function
a) Make helper column
b) =trim(M2) and enter
c) Copy paste values

Use Case for Flash Fill:

If you have full names in the "Customer Contact" column and want to split the first name and last name into two separate columns.

Using Flash Fill in Excel for the Web:

To do this, first type the last name of the first record. Then, under the "Data" tab, click on the "Flash Fill" option.

This will automatically fill the remaining cells with the last names extracted from the original column. You can do the same for the first names as well.

More excel features for cleaning data

I. Flash fill to combine names
   A. New helper column
   B. Type in Jonas Bergulfsen, then hit enter, will see it auto fill
II. Flash fill to modify naming in columns
III. Need to use text to columns feature to split names into two columns
   A. Text to columns can split text into one or more columns into separate component parts
      1. Select data, then text to columns, ensure delimited is selected and space is selected as delimiter, then select where destination is
      2. Can also do this with functions
         a) =left(A2,search(“ “)

Summary and Highlights

In this lesson, you have learned the following information:

- It's important to remove any duplicated or inaccurate data, and it's important to remove any empty rows in your dataset.
- There are several other types of data inconsistency that you may need to resolve, in order to properly clean your data:
1. Change the case of text
2. Fix date formatting errors
3. Trim whitepace from your data
- You can use the Flash Fill and Text to Columns features in Excel to manipulate and standardize your data, and functions can also be used to help manipulate and standardize your data.

Intro to analyzing data using spreadsheets

I. How to shape our data
    A. How big is dataset
    B. What type of filtering is required to find info
    C. How should be sorted
    D. What calculations should be made
II. FIltering and sorting data
    A. Alphabetically or numerically
III. Using functions in data analysis
    A. One key benefit of using spreadsheet is ability to use functions
        1. Mathematical, statistical, logical, financial, date and time
IV. Analyzing data in a table
    A. Can easily calculate columns
    B. Formating in table benefits
        1. Automatic calculations w
        2. Column headings never disappear
        3. Banded rows to make reading easier
        4. Auto expand when adding new rows
    C. Pivot tables and pivot charts
        1. Nice way to show only info that is required
        2. Pivot charts allow us to visually process data and grasp info quicker
        3. Advantages
            a) Manipulate data w/o formulas

b) Quickly summarize large data sets
c) Show data in engaging charts and graphs

FIltering and sorting data in excel

I. Data>Filter
    A. If format as table, filter is auto added
    B. Want to clear all filters, use clear button on data tab
    C. Can use custom filters
        1. Greater than or less than “X”
II. Sorting
    A. Different ways:
        1. Text-based data alphabetically
        2. Number based data numerically
        3. Date based data chronologically
    B. Data>Sort
        1. Can add another level to make it sort by one column then another

Useful functions for data analysis

I. IF Functions
    A. One of the most used logical functions
    B. Logically compare a value against criteria, returns result based on true or false comparisons
    C. If something is true then return a value or return something else
    D. Use when only two conditions exist
    E. Can use multiple ‘nested’ IFs when several conditions exist
        1. =if(E2>700,”Large”,if(E2>4000,”medium”,if(E2>0,”Small”))
            a) Supports up to 64 nested ifs but not a best practice
                (1) Can be hard to manage and understand
    F. New function called “IFS” eliminat4es need for multiple “if’s”
        1. Use =IFS instead of always writing “IF”
    G. Can combine with conditional formatting
        1. Conditional formatting. Only values that have “Good” as an example
II. Count IF
    A. Common statistical excel function
    B. Counts number of cell values that meet specified criteria
    C. COUNTIFS function eliminates need for multiple nested ‘countif’s
III. Sum IF
    A. Values in a given range that match specified criteria

B. Can use wild cards
C. Also have SUMIFS that removes need for multiple nested sumif's

VLOOKUP and HLOOKUP functions

I. Vlookup allows you to find data referenced in a lookup table
II. Vertical lookup- find by column
III. Hlookup is horizontal lookup - find by row
IV. Works by using common shared key
    A. =vlookup(B3,A2:B12,2,false)
        1. B3 is lookup value
        2. A2-B12 is range
        3. 2 is Lookup column number (# of column you are looking for)
        4. False - optional to denote if it needs to be exact (false) or approximate (true)
            a) Defaults to true
        5. Put column containing value we are looking for in the left most column, which is required for vlookup
        6. F4 automatically adds $ to the column and rows
    B. HLooKUP
        1. Looks for data in rows, rather than columns, returns value in top row in table
        2. Use HLOOKUP if comparison values were situated in a row along top of a data table
        3. Use VLOOKUP if your comparison values were located in a column to the left of the data
            a) Used much more frequently due to context of data
    C. Xlookup
        1. Improved and combined version of vlookup and hlookup
        2. Works in any direction
        3. Uses separate lookup array and return array values

Differentiation among VLOOKUP, HLOOKUP, and XLOOKUP:

- **VLOOKUP**
    - Searches for a value in the first column of a table and returns a value in the same row from a specified column.
    - Limited to vertical searches and can only search left to right.
    - Prone to errors if columns are added or removed.
- **HLOOKUP**
    - Similar to VLOOKUP but searches horizontally.
    - Limited to horizontal searches and can only search top to bottom.
- **XLOOKUP**
    - More flexible, can search in both vertical and horizontal ranges.

  - Can search in either direction and provides more robust error handling.
  - Replaces both `VLOOKUP` and `HLOOKUP` with enhanced functionality.

Importance and Uses:

- **Data Retrieval:** Efficient for retrieving data from large datasets.

Summary and Highlights

In this lesson, you have learned the following information:

Before shaping your data, you need to visualize the final output, and ask yourself the following questions:

- How big is the dataset?
- What type of filtering is required to find the necessary information?
- How should the data be sorted?
- What type of calculations are needed?

There are several advantages to formatting your data as a table:

- Automatic calculations even when filtering
- Column headings never disappear
- Banded rows to make reading easier
- Tables will automatically expand when adding new rows

The most basic way of shaping your data is to sort and filter it:

- Sorting data helps you to organize it by a specified criteria, such as numerically, alphabetically, or chronologically.
- Filtering our data makes it easier to control what data is displayed and what is hidden, based on filtered fields.

Excel Functions:

- Functions in Excel are arranged into multiple categories; including mathematical, statistical, logical, financial, and date and time-based.
- Common functions for a data analyst include IF, IFS, COUNTIF, SUMIF, VLOOKUP, HLOOKUP

Introduction to creating pivot tables in excel

I. Pivot tables provide a simple and quickly to summarize and analyze data, to observe trends and patterns in data and make data comparisons
    A. Dynamic, can incorporate more data as added
II. Pivot table checklist
    A. Format data as a table
    B. Ensure column headings are correct and only one header row
    C. Remove blank rows, column s and cells
    D. Ensure value fields are formatted as numbers
    E. Ensure date fields are formatted as dates and not text
III. Select insert>Pivot table
    A. New worksheet is default and most common
    B. Fields, items and sets, can tell it to add a formula and will add to pivot table pane

Pivot table features

I. Recommended pivot tables
    A. Automatically suggests recommended pivot table as it relates to columns that we have highlighted
II. Pivot table filtering
III. Slicers
    A. On screen graphical filters
        1. Display current filter state
    B. Pivot table analyze tab > insert slicer
        1. Select cell in pivot table first
        2. Can select multi and clear all filters buttons
        3. Selecting items that will be shown on the pivot table
IV. Timelines
    A. Enables to filter on date related data
        1. Add timeline from pivot analyze or insert tab
        2. Select cell in pivot table first
        3. Order date field as
        4. Slicer will show up and it can filter by years, quarters, etc.

Summary and Highlights

In this lesson, you have learned the following information:

Pivot Tables:

- To obtain usable and presentable insights into your data you need to use Pivot Tables.
- Pivot tables provide a simple and quick way to summarize and analyze data, to observe trends and patterns in your data and to make comparisons of your data.
- Pivot tables are dynamic, so as you change and add data to the original dataset on which the pivot table is based, the analysis and summary information changes too.
- A Data Analyst can use pivot tables to draw useful and relevant conclusions about, and create insights into, an organization's data in order to present those insights to interested parties within the company.

Use this Pivot Table checklist to ensure your data is in a fit state to make a Pivot Table:

- Format your data as a table for best results.
- Ensure column headings are correct, and there is only one header row, as these column headings become the field names in a Pivot Table.
- Remove any blank rows and columns, and try to eliminate blank cells also.

- Ensure value fields are formatted as numbers, and not text, and ensure date fields are formatted as dates, and not text.

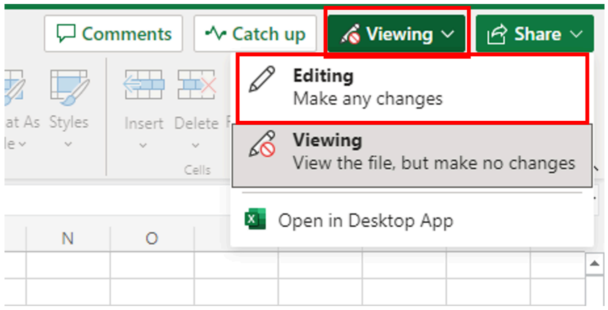
Arranging Pivot Tables with Filters and Recommended Tables:

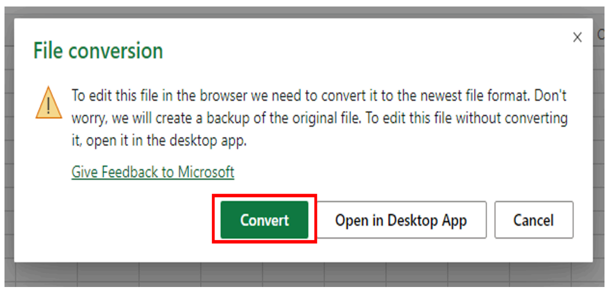
- You use the Pivot Table Fields pane to add and arrange data fields in your pivot table.
- Recommended Pivot Tables are a list of suggested different combinations of data that could be used when creating a Pivot Table, based on the data selected in the worksheet.

Filters and Slicers:

- Slicers are on-screen graphical filter objects that enable you to filter your data using buttons, which makes it easier to perform quick filtering of your pivot table data.
- Timelines are another type of filter tool that enable you to filter specifically on date-related data in your pivot table. This is a much quicker and more effective way of dynamically filtering by date, rather than having to create and adjust filters on your date columns.

asks to perform:

1. **Save the CSV file as an XLSX file:** Change the ‘Viewing’ in the ToolTip to ‘Editing’inorder to save the file as an XLSX file. The file is converted when you click ‘Convert’ in the prompt.




2. **Column widths:** Sort out the widths of all columns so that the data is clearly visible in all cells.
3. **Empty rows:** Use the Filter feature to look for blanks and remove all empty rows from the data.
4. **Duplicate records:** Use either the Conditional Formatting or Remove Duplicates feature to look for and remove any duplicated records from the data.
5. **Spelling:** The original source file data has not been checked for errors in the spelling. Check for spelling mistakes in the data and fix them.
6. **Whitespace:** Use the Find and Replace feature to remove all double-spaces from the data.

7. **Department names:** When the data was converted from its data source, the department names (see correct list below) didn’t import correctly and they are now split over two columns in the data. Use Flash Fill to reduce the department names to just one column, and then remove any unnecessary columns.

Department	Department
Board of Elections	Economic Development

Circuit Court	Environmental Protection
Community Engagement Cluster	Finance
Community Use of Public Facilities	Fire and Rescue
Consumer Protection	General Services
Correction and Rehabilitation	Health and Human Services
County Executives Office	

8. **Download your workbook:** Use‘Save As’and select‘Download a copy’ to download your completed workbook as **Montgomery_Fleet_Equipment_Inventory_FA_PART_1_END.XLSX**.